Notes: Couts, Gonçalves, and Rossi (RFS, 2024)
Unsmoothing Returns of Illiquid Funds

Contents

1 Big picture

- **Question.** How should investors measure risk exposures and alphas for funds that invest in illiquid assets and therefore report smoothed returns?
- **Core problem.** Observed fund returns are weighted averages of current and lagged economic returns. This smoothing creates spurious autocorrelation, lowers measured volatility, and can overstate Sharpe ratios and alphas.
- **Existing solution.** Traditional 1-step unsmoothing methods, such as Geltner for real estate and Getmansky–Lo–Makarov for hedge funds, remove much fund-level autocorrelation.
- **New insight.** Even after 1-step unsmoothing, returns of funds with similar strategies still have autocorrelated aggregate components. Systematic risk remains smoothed.
- **Main method.** The paper proposes a 3-step unsmoothing procedure that separately unsmooths the aggregate component of similar funds and the fund-specific component relative to that aggregate.
- **Hedge fund evidence.** In 5,069 U.S.-dollar hedge funds from 1995 to 2017, 1-step unsmoothing removes most fund-level autocorrelation but leaves high autocorrelation in strategy-level returns. The 3-step method largely removes both.
- **Risk measurement effect.** The 3-step method raises factor-model R^2s for illiquid strategies, raises measured systematic risk exposures, and lowers average alphas and alpha t-statistics.
- **Private real estate evidence.** Applying a 3-step AR unsmoothing method to private commercial real estate funds materially increases volatility and R^2 and reduces the average alpha toward zero.

2 Data and measurement

2.1 Hedge fund sample

- The hedge fund sample runs from January 1995 to December 2017.
- The baseline sample keeps U.S. dollar funds reporting net-of-fees returns.
- Funds must have at least 36 uninterrupted monthly observations and reach $5 million in AUM at some point in the sample.
- The final hedge fund sample contains 5,069 funds across ten strategies.
- The strategies are sorted by first-order average fund-level autocorrelation: relative value, event driven, multistrategy, emerging markets, sector, long only, long-short, market neutral, global macro, and CTA.

Interpretation. The strategy ranking gives an empirical proxy for illiquidity.

2.2 Return definitions and risk measures

- **Observed returns** are the reported fund returns.
- **1-step unsmoothed returns** are obtained using standard fund-by-fund moving-average unsmoothing.
- **3-step unsmoothed returns** separately account for smoothing in the aggregate component and in the fund-specific relative component.
- The paper compares autocorrelations, volatility, Sharpe ratios, factor-model R^2, alphas, alpha t-statistics, and the fraction of significant alphas across these return definitions.

Interpretation. If unsmoothing works, autocorrelation should not survive at either the fund level or the strategy-aggregate level.

2.3 Private commercial real estate sample

- The private CRE sample comes from NCREIF.
- It contains 66 U.S. private commercial real estate funds.
- The sample runs from Q1 1994 to Q4 2017 and requires at least 36 quarterly observations.
- Risk models use either one public real estate factor or a two-factor model with public real estate and public equity factors.

Interpretation. The CRE application shows that the same measurement issue arises when appraisal smoothing creates strongly autoregressive returns.

3 Models and empirical specifications

3.1 One-step moving-average unsmoothing

The standard unsmoothing model assumes that observed fund returns are smoothed versions of economic returns:

$$R^o_{j,t} = \theta_j^{(0)} R_{j,t} + \theta_j^{(1)} R_{j,t-1} + \cdots + \theta_j^{(H)} R_{j,t-H}. \tag{1}$$

If $R_{j,t} = \mu_j + \eta_{j,t}$, then

$$R^o_{j,t} = \mu_j + \sum_{h=0}^{H} \theta_j^{(h)} \eta_{j,t-h}, \qquad \sum_{h=0}^{H} \theta_j^{(h)} = 1. \tag{2}$$

Interpretation. The 1-step method estimates smoothing weights and extracts residuals as economic returns. It is correct if the same smoothing process applies to all relevant return components.

3.2 Three-step moving-average unsmoothing

Define the aggregate return and relative return as

$$R_t = \sum_j w_j R_{j,t}, \qquad \widetilde{R}_{j,t} = R_{j,t} - R_t. \tag{3}$$

The 3-step model allows different smoothing weights for the relative and aggregate components:

$$R^o_{j,t} = \sum_{h=0}^{H} \phi_j^{(h)} \widetilde{R}_{j,t-h} + \sum_{h=0}^{L} \pi_j^{(h)} R_{t-h}, \tag{4}$$

$$R^o_{j,t} = \mu_j + \sum_{h=0}^{H} \phi_j^{(h)} \widetilde{\eta}_{j,t-h} + \sum_{h=0}^{L} \pi_j^{(h)} \eta_{t-h}, \qquad \sum_h \phi_j^{(h)} = \sum_h \pi_j^{(h)} = 1. \tag{5}$$

Interpretation. The standard 1-step method is the special case in which aggregate and relative components have the same smoothing weights.

3.3 Failure of 1-step unsmoothing

If the true process is 3-step but the econometrician uses 1-step unsmoothing, the recovered return contains an extra lagged aggregate-error component:

$$R^{1s}_{j,t} = \mu_j + \eta_{j,t} + \sum_h \lambda_j^{(h)} \epsilon_{j,t-h}. \tag{6}$$

At the aggregate level,

$$R_t^{1s} \approx \mu + \eta_t + \sum_h \lambda^{(h)}(1-b)\eta_{t-h}. \tag{7}$$

Interpretation. The misspecification is hard to see at the fund level because idiosyncratic shocks dominate, but it is easy to see after aggregation because aggregation emphasizes the systematic component.

3.4 Factor models and private CRE autoregressive unsmoothing

For each return definition, the paper estimates factor models of the form

$$R_{j,t}^u = \alpha_j + \beta_j' f_t + \varepsilon_{j,t}. \tag{8}$$

For private CRE funds, the autoregressive unsmoothing framework is

$$R_{j,t}^o = \theta_j^{(0)} R_{j,t} + \sum_{h=1}^{H} \theta_j^{(h)} R_{j,t-h}^o. \tag{9}$$

Interpretation. If unsmoothing raises betas and R^2, alpha should fall. A large alpha reduction means earlier alpha was compensation for hidden systematic risk.

4 Identification and empirical design

- The paper identifies a measurement problem using internal consistency restrictions, not a causal policy shock.
- If 1-step unsmoothing recovers true economic returns, both fund-level and aggregate-level autocorrelations should disappear.
- Empirically, 1-step unsmoothing removes fund-level autocorrelation but not aggregate-level autocorrelation.
- The paper uses two-group unsmoothing to address mechanical-overlap concerns.
- It also uses beta-sorted portfolios, simulations, out-of-sample alpha portfolio tests, and a private CRE application.

Interpretation. Multiple diagnostics point to the same conclusion: the 1-step method understates systematic risk for illiquid funds.

5 Tables and figures

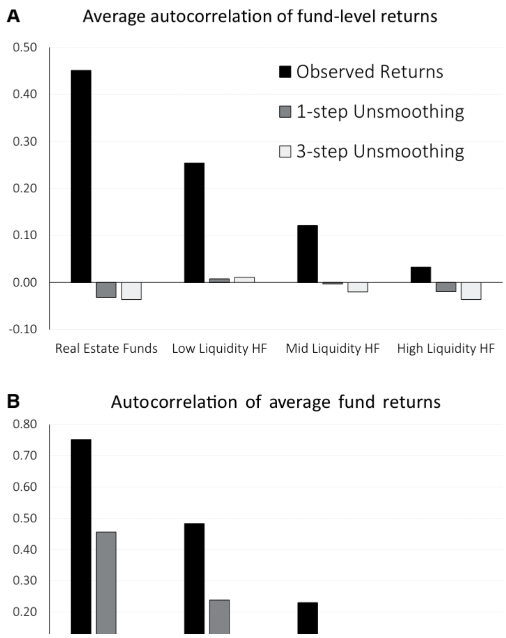
The original visuals below are directly cropped from the uploaded paper and grouped to save space.

5.1 Figure 1 and Table 1: the problem and the sample

Read:

- Figure 1 shows that 1-step unsmoothing removes fund-level autocorrelation but leaves aggregate autocorrelation. The 3-step method largely removes both.
- Table 1 shows 5,069 hedge funds. Relative value, event driven, and multistrategy funds have the highest first-order autocorrelations.

Economic message. Smoothing is strongest in illiquid strategies, and the key failure of 1-step unsmoothing is at the aggregate systematic component.



(a) Figure 1: autocorrelation

Table 1
Hedge fund strategies and summary statistics

Hedge fund	**Sample size**		**Fund level**			
Strategies	N	$\overline{T}$	Cor_1	σ	$\mathbb{E}[r]$	$\mathbb{E}[r]/\sigma$
Relative value	670	85	0.29	8.5%	3.4%	0.40
Event driven	433	96	0.25	9.5%	4.9%	0.51
Multistrategy	199	97	0.22	9.5%	3.4%	0.36
Emerging mkts	657	92	0.18	17.7%	3.4%	0.19
Sector	318	90	0.13	15.1%	2.9%	0.20
Long only	455	102	0.11	15.3%	5.1%	0.33
Long-short	965	92	0.10	14.2%	3.3%	0.23
Market neutral	201	79	0.09	6.9%	1.5%	0.21
Global macro	212	93	0.06	13.4%	3.1%	0.23
CTA	959	94	0.00	15.5%	2.4%	0.15

The table reports the total number of hedge funds (N), the average number of months per hedge fund ($\overline{T}$), and other average fund-level statistics for hedge fund returns by strategy, with strategies sorted based on the first-order (average fund-level) return autocorrelation. All statistics are based on observed returns. The sample goes from January 1995 to December 2017 and is restricted to U.S. dollar funds that report net-of-fees returns, have at least 36 uninterrupted monthly observations, and reach $5 million in AUM at some point in the sample, with fund observations included only after reaching the $5 million AUM threshold for the first time. See Subsection 2.1 for further empirical details.

(b) Table 1: summary statistics

5.2 Tables 2 and 3: fund-level versus aggregate autocorrelation

Read:

- Table 2 shows that 1-step unsmoothing drives fund-level autocorrelations close to zero.
- Table 3 shows that aggregate strategy returns remain autocorrelated after 1-step unsmoothing but not after 3-step unsmoothing.

Economic message. The 1-step method solves the idiosyncratic smoothing problem but leaves the common risk component smoothed.

5.3 Tables 4 and 5: systematic-risk portfolios and simulations

Read:

- Table 4 shows beta-sorted portfolios remain autocorrelated after 1-step unsmoothing but are close to zero after 3-step unsmoothing.
- Table 5 shows that 3-step unsmoothing matters when aggregate and fund-specific components are smoothed differently.

Economic message. The 3-step method is needed exactly when aggregate and idiosyncratic components are smoothed differently.

5.4 Figures 2 and 3: risk, alpha, and out-of-sample performance

Read:

- Figure 2 shows that 3-step unsmoothing raises volatility and R^2 and lowers alpha for illiquid strategies.
- Figure 3 shows that 3-step alpha estimates have better out-of-sample meaning, especially for illiquid funds.

Table 2
Autocorrelations of hedge fund returns

Hedge fund Strategies	Observed returns Cor_1	Cor_2	Cor_3	1-step unsmoothing Cor_1	Cor_2	Cor_3	3-step unsmoothing Cor_1	Cor_2	Cor_3	Two group 3-step unsmoothing Cor_1	Cor_2	Cor_3
Relative value	0.29	0.18	0.19	0.02	0.04	0.09	0.04	−0.03	0.10	0.03	−0.02	0.07
	(61%)	(38%)	(35%)	(3%)	(3%)	(4%)	(17%)	(7%)	(17%)	(14%)	(6%)	(8%)
Event driven	0.25	0.13	0.10	0.01	0.01	0.01	−0.01	0.01	0.01	−0.02	−0.01	0.01
	(62%)	(37%)	(32%)	(2%)	(2%)	(3%)	(11%)	(7%)	(9%)	(10%)	(4%)	(9%)
Multistrategy	0.22	0.11	0.07	−0.01	−0.01	0.00	0.00	−0.03	0.00	0.00	−0.04	−0.02
	(57%)	(37%)	(25%)	(1%)	(2%)	(1%)	(11%)	(5%)	(6%)	(10%)	(4%)	(2%)
Emerging mkts	0.18	0.07	0.06	0.00	−0.01	0.00	−0.04	0.00	0.03	−0.05	0.00	0.02
	(47%)	(24%)	(16%)	(1%)	(0%)	(1%)	(7%)	(8%)	(8%)	(5%)	(10%)	(6%)
Sector	0.13	0.05	0.02	0.00	−0.01	−0.02	−0.01	0.01	−0.04	−0.02	0.01	−0.02
	(36%)	(19%)	(10%)	(1%)	(0%)	(1%)	(7%)	(5%)	(2%)	(5%)	(4%)	(1%)
Long only	0.11	0.03	0.04	0.00	−0.01	−0.01	−0.03	−0.01	−0.01	−0.01	−0.01	−0.01
	(37%)	(10%)	(10%)	(1%)	(0%)	(0%)	(6%)	(3%)	(2%)	(6%)	(4%)	(3%)
Long-short	0.10	0.03	0.04	−0.01	−0.02	−0.01	−0.01	−0.02	−0.01	−0.01	−0.01	−0.02
	(30%)	(14%)	(12%)	(1%)	(1%)	(2%)	(5%)	(5%)	(3%)	(5%)	(5%)	(3%)
Market neutral	0.09	0.03	0.02	−0.01	−0.02	−0.02	−0.01	−0.03	−0.04	0.00	−0.06	−0.04
	(27%)	(13%)	(14%)	(1%)	(1%)	(1%)	(4%)	(3%)	(2%)	(1%)	(0%)	(4%)
Global macro	0.06	0.00	−0.02	−0.01	−0.02	−0.03	−0.03	−0.01	−0.03	−0.04	−0.01	−0.02
	(23%)	(11%)	(7%)	(2%)	(2%)	(1%)	(3%)	(5%)	(2%)	(1%)	(3%)	(2%)
CTA	0.00	0.00	−0.02	−0.03	−0.02	−0.03	−0.04	−0.02	−0.04	−0.04	−0.02	−0.03
	(9%)	(9%)	(5%)	(1%)	(1%)	(0%)	(1%)	(4%)	(3%)	(1%)	(3%)	(2%)

The table reports average fund-level autocorrelations (from 1 to 3 months) for hedge fund returns by strategy, with strategies sorted based on the first-order (average fund-level) observed return autocorrelation. Reported autocorrelations are based on observed returns, 1-step unsmoothed returns (as in Getmansky, Lo, and Makarov 2004), and 3-step unsmoothed returns. For the columns under "Two group 3-step unsmoothing," we rank funds based on the alphabetic order of their names within strategy at the year of the fund inception, create two groups based on even versus odd ranks, and use the aggregate index from the second (first) group to unsmooth the returns of the first (second) group, with the table reporting results based on the unsmoothed returns from the first group (the results from the second group are similar). The numbers in parentheses reflect the fraction of funds with the respective autocorrelation being significant at 10% level. The sample goes from January 1995 to December 2017 and is restricted to U.S. dollar funds that report net-of-fees returns, have at least 36 uninterrupted monthly observations, and reach $5 million in AUM at some point in the sample, with fund observations included only after reaching the $5 million AUM threshold for the first time. See Section 1 for unsmoothing methods and Subsection 2.1 for further empirical details.

(a) Table 2: fund-level autocorrelations

Table 3
Autocorrelations of aggregated hedge fund returns

Hedge fund Strategies	Observed returns Cor_1	Cor_2	Cor_3	1-step unsmoothing Cor_1	Cor_2	Cor_3	3-step unsmoothing Cor_1	Cor_2	Cor_3	Two group 3-step unsmoothing Cor_1	Cor_2	Cor_3
Relative value	0.51	0.28	0.14	0.27	0.12	0.04	0.01	0.03	0.10	0.00	0.05	0.08
	(0%)	(0%)	(2%)	(0%)	(5%)	(51%)	(85%)	(57%)	(9%)	(97%)	(42%)	(17%)
Event driven	0.46	0.25	0.18	0.21	0.09	0.08	0.03	0.03	0.05	−0.02	0.01	0.01
	(0%)	(0%)	(0%)	(0%)	(15%)	(20%)	(62%)	(68%)	(46%)	(70%)	(82%)	(83%)
Multistrategy	0.48	0.34	0.24	0.23	0.15	0.11	0.06	0.03	0.05	0.02	−0.02	0.01
	(0%)	(0%)	(0%)	(0%)	(1%)	(8%)	(33%)	(62%)	(41%)	(78%)	(72%)	(83%)
Emerging mkts	0.32	0.11	0.06	0.14	0.04	0.02	0.01	0.02	0.03	−0.02	0.03	0.03
	(0%)	(7%)	(30%)	(2%)	(48%)	(70%)	(93%)	(69%)	(64%)	(74%)	(66%)	(67%)
Sector	0.21	0.05	0.04	0.07	−0.01	0.03	0.02	0.00	0.00	0.04	0.04	−0.05
	(0%)	(39%)	(49%)	(28%)	(87%)	(64%)	(74%)	(96%)	(97%)	(55%)	(48%)	(38%)
Long Only	0.22	0.05	0.04	0.09	0.02	0.00	0.00	0.00	0.01	0.02	−0.03	0.00
	(0%)	(41%)	(56%)	(12%)	(71%)	(96%)	(99%)	(100%)	(92%)	(76%)	(64%)	(94%)
Long-short	0.23	0.11	0.08	0.10	0.06	0.04	0.00	0.00	0.02	−0.02	−0.02	0.00
	(0%)	(7%)	(18%)	(12%)	(36%)	(52%)	(98%)	(95%)	(74%)	(70%)	(80%)	(97%)
Market neutral	0.18	0.10	0.06	0.09	0.07	0.03	0.03	0.04	0.04	0.01	−0.02	0.13
	(0%)	(9%)	(30%)	(14%)	(25%)	(61%)	(57%)	(54%)	(47%)	(92%)	(73%)	(4%)
Global macro	0.05	−0.03	0.00	−0.01	−0.03	0.01	0.01	−0.01	0.01	0.01	−0.08	0.04
	(37%)	(63%)	(99%)	(93%)	(64%)	(90%)	(87%)	(84%)	(93%)	(87%)	(18%)	(47%)
CTA	−0.02	−0.03	0.00	−0.04	−0.01	0.02	−0.01	0.03	0.01	−0.05	0.00	0.02
	(71%)	(61%)	(98%)	(54%)	(91%)	(70%)	(88%)	(57%)	(86%)	(37%)	(98%)	(79%)

The table reports autocorrelations (from 1 to 3 months) for returns of each hedge fund strategy index (i.e. equal-weighted portfolio of all funds following the given strategy), with strategies sorted based on the first-orde (average fund-level) observed return autocorrelation. Reported autocorrelations are based on observed returns 1-step unsmoothed returns (as in Getmansky, Lo, and Makarov 2004), and 3-step unsmoothed returns. For the columns under "Two Group 3-step Unsmoothing", we rank funds based on the alphabetic order of their name within strategy at the year of the fund inception, create two groups based on even versus odd ranks, and use the aggregate index from the second (first) group to unsmooth the returns of the first (second) group, with the table reporting results based on the unsmoothed returns from the first group (the results from the second group ar similar). The numbers in parentheses reflect the p-value for the test of whether the respective autocorrelation differs from zero. The sample goes from January 1995 to December 2017 and is restricted to U.S. dollar fund that report net-of-fees returns, have at least 36 uninterrupted monthly observations, and reach $5 million in AUM at some point in the sample, with fund observations included only after reaching the $5 million AUM threshold for the first time. See Section 1 for unsmoothing methods and Subsection 2.1 for further empirical details.

(b) Table 3: aggregate autocorrelations

Table 4
Autocorrelations of β-sorted hedge fund portfolio returns

Hedge fund Strategies	Observed returns Low β	Mid β	High β	1-step Unsmoothing Low β	Mid β	High β	3-step Unsmoothing Low β	Mid β	High β
Market	0.15	0.33	0.21	0.06	0.16	0.10	−0.01	0.05	0.03
	(1%)	(0%)	(0%)	(30%)	(1%)	(11%)	(86%)	(44%)	(66%)
Size spread	0.08	0.27	0.24	0.00	0.13	0.11	−0.05	0.01	0.04
	(18%)	(0%)	(0%)	(94%)	(4%)	(7%)	(39%)	(92%)	(51%)
Emerging market	0.20	0.30	0.23	0.12	0.16	0.08	−0.03	0.04	0.02
	(0%)	(0%)	(0%)	(6%)	(1%)	(17%)	(67%)	(50%)	(73%)
Bond market	0.16	0.32	0.25	0.08	0.16	0.10	0.02	0.03	0.02
	(1%)	(0%)	(0%)	(20%)	(1%)	(8%)	(80%)	(64%)	(79%)
Credit spread	0.35	0.26	0.15	0.20	0.12	0.05	0.12	0.02	−0.04
	(0%)	(0%)	(1%)	(0%)	(4%)	(38%)	(5%)	(80%)	(50%)
Bond trend following	0.23	0.33	0.13	0.10	0.17	0.05	0.00	0.04	0.00
	(0%)	(0%)	(3%)	(11%)	(1%)	(39%)	(100%)	(49%)	(97%)
Currency trend following	0.35	0.28	0.08	0.18	0.14	0.01	0.07	0.02	−0.03
	(0%)	(0%)	(18%)	(0%)	(3%)	(85%)	(25%)	(77%)	(58%)
Commodity trend following	0.36	0.33	0.04	0.20	0.16	−0.03	0.11	0.03	−0.08
	(0%)	(0%)	(54%)	(0%)	(1%)	(65%)	(7%)	(61%)	(19%)

The table reports first-order return autocorrelations of 24 portfolios. To construct the portfolios, we start by estimating the Fung and Hsieh (2001) factor model betas for each fund using observed returns. We then sort all funds into three groups based on their exposures to each of the eight factors in the model, which yields $3 \times 8 = 24$ portfolios. Autocorrelations are based on observed returns, 1-step unsmoothed returns (as in Getmansky, Lo, and Makarov 2004), and 3-step unsmoothed returns, with fund-level unsmoothed returns being the same ones used in Table 3. The numbers in parentheses reflect the p-value for the test of whether the respective autocorrelation differs from zero. The sample goes from January 1995 to December 2017 and is restricted to U.S. dollar funds that report net-of-fees returns, have at least 36 uninterrupted monthly observations, and reach $5 million in AUM at some point in the sample, with fund observations included only after reaching the $5 million AUM threshold for the first time. See Section 1 for unsmoothing methods and Subsection 2.1 for further empirical details.

First, within each strategy, we divide funds into two groups based on the

(a) Table 4: beta-sorted portfolios

Table 5
Unsmoothing techniques in simulations

A. MA(1) with 1-factor model

	$\phi^{(1)} = \pi^{(1)}$		$\phi^{(1)} < \pi^{(1)}$		$\phi^{(1)} > \pi^{(1)}$	
$\phi^{(1)}$	0.30		0.20		0.40	
$\pi^{(1)}$	0.30		0.40		0.20	
$Cor_1(R_o)$	0.35		0.31		0.32	
$Cor_1(R_{1s})$	0.00		0.00		0.00	
$Cor_1(R_{3s})$	−0.01		−0.01		−0.02	
$Cor_1(\overline{R}_o)$	0.36		0.46		0.24	
$Cor_1(\overline{R}_{1s})$	0.01		0.24		−0.14	
$Cor_1(\overline{R}_{3s})$	0.00		−0.01		0.00	
	Standard	+Dimson	Standard	+Dimson	Standard	+Dimson
$\widehat{\alpha}_o$	2.1%	0.0%	2.8%	0.0%	1.4%	0.0%
$\widehat{\alpha}_{1s}$	0.1%	0.0%	1.3%	−0.4%	−0.7%	0.3%
$\widehat{\alpha}_{3s}$	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
$\lvert\widehat{\alpha}_o\rvert$	2.3%	0.6%	3.0%	0.6%	1.6%	0.6%
$\lvert\widehat{\alpha}_{1s}\rvert$	0.5%	0.7%	1.4%	0.8%	0.9%	0.7%
$\lvert\widehat{\alpha}_{3s}\rvert$	0.3%	0.7%	0.4%	0.8%	0.4%	0.8%
$100 \times \widehat{\alpha}_o^2$	0.074	0.006	0.124	0.007	0.042	0.006
$100 \times \widehat{\alpha}_{1s}^2$	0.004	0.009	0.029	0.014	0.014	0.008
$100 \times \widehat{\alpha}_{3s}^2$	0.002	0.011	0.003	0.011	0.002	0.011

The table reports autocorrelations and alphas in simulations. In panel A, we simulate returns for a panel of 670 funds over a 85-month period. The monthly economic returns of each fund j satisfy $R_{j,t} = \alpha_j + \beta_j \cdot f_t + \varepsilon_{j,t}$, where $\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2)$, $\beta_j \sim N(1, \sigma_\beta^2)$, $f_t \sim N(\mu_f, \sigma_f^2)$, and $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. We then smooth these returns according to the underlying smoothing process of our 3-step method (i.e., Equation (5)) with $H = L = 1$ (i.e., MA(1) smoothing), $\phi_j^{(1)} = \phi^{(1)}$, and $\pi_j^{(1)} = \pi^{(1)}$ for simplicity. Finally, we estimate economic returns for each fund in the panel using the 1- and 3-step unsmoothing methods and study the properties of observed returns, 1-step unsmoothed returns, and 3-step unsmoothed returns. Columns with "+Dimson" apply the Dimson (1979) method (with one lag) to the given return measure. We report the average results obtained from 1,000 simulations of this panel of funds. The first column shows results for a specification in which the aggregate and fund-specific components of returns are smoothed with the same intensity ($\phi^{(1)} = \pi^{(1)} = 0.3$), the second column considers a specification in which the fund-level component of returns is smoothed less than the aggregate-level component ($\phi^{(1)} = 0.2$ and $\pi^{(1)} = 0.4$), and the third column considers an alternative scenario in which the

(b) Table 5A: MA(1) simulation

Economic message. Some illiquid-fund alpha was compensation for systematic risk hidden by smoothing.

5.5 Figures 4–6: strategy-level risk and factor decomposition

Read:

- Figure 4 shows that 3-step unsmoothing increases volatility and R^2 and reduces alpha measures across

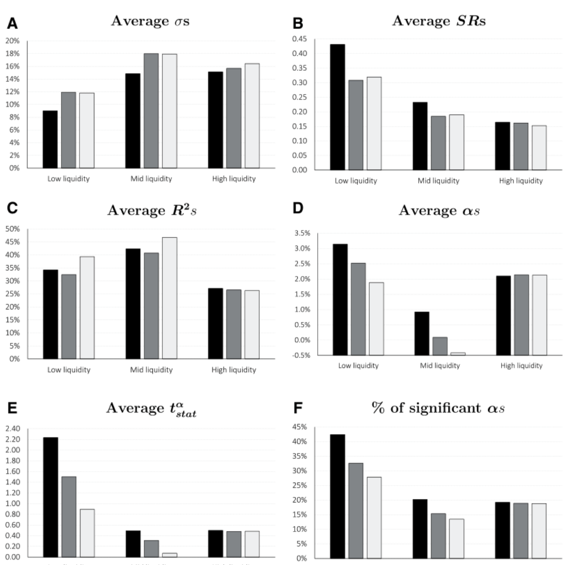
Table 5
(Continued)

B. MA(3) with 8-factor model

	$T = 85$ Months			$T = 170$ Months		
$Cor_1(R_o)$		0.13			0.14	
$Cor_1(R_{1s})$		−0.01			0.00	
$Cor_1(R_{3s})$		−0.01			−0.01	
$Cor_1(\overline{R}_o)$		0.46			0.47	
$Cor_1(\overline{R}_{1s})$		0.28			0.28	
$Cor_1(\overline{R}_{3s})$		0.00			0.01	
	Standard	+Dimson	+CDimson	Standard	+Dimson	+CDimson
$\widehat{\alpha}_o$	0.7%	0.0%	0.1%	0.7%	0.0%	0.2%
$\widehat{\alpha}_{1s}$	0.4%	0.0%	0.1%	0.4%	0.1%	0.1%
$\widehat{\alpha}_{3s}$	0.0%	0.0%	−0.3%	0.0%	0.0%	−0.2%
$\lvert\widehat{\alpha}_o\rvert$	1.1%	2.4%	1.3%	1.0%	1.7%	1.0%
$\lvert\widehat{\alpha}_{1s}\rvert$	0.9%	2.6%	1.4%	0.7%	1.9%	1.0%
$\lvert\widehat{\alpha}_{3s}\rvert$	0.8%	2.7%	1.3%	0.6%	1.9%	0.9%
$100 \times \widehat{\alpha}_o^2$	0.024	0.115	0.039	0.019	0.074	0.026
$100 \times \widehat{\alpha}_{1s}^2$	0.016	0.141	0.043	0.011	0.091	0.027
$100 \times \widehat{\alpha}_{3s}^2$	0.012	0.147	0.039	0.008	0.095	0.027

The table reports autocorrelations and alphas in simulations. In panel B, we simulate returns for a panel of 670 funds over a 85-month period or a 170-month period. The monthly economic returns of each fund j satisfy $R_{j,t} = \alpha_j + \beta_j' f_t + \varepsilon_{j,t}$ with β_j representing a vector of eight fund-specific risk exposures and f_t representing a vector of eight risk factors. Simulated fund-level parameters (i.e., β_j and the vectors of smoothing parameters, π_j and ϕ_j) are bootstrapped with replacement from the corresponding joint empirical distribution of parameters based on relative value funds estimated under the 3-step method with three lags. The panel of factor returns is common to all funds within a given simulation run and is bootstrapped with replacement from the time series of the 8 FH factors. For simplicity, "true alpha" (α_j) is set to zero for all funds. Finally, the standard deviation of $\varepsilon_{j,t}$ is set to 2.2%, which ensures that the smoothed simulated fund returns have the same average volatility of the reported returns of relative value funds. Finally, we estimate economic returns for each fund in the panel using the 1- and 3-step unsmoothing methods and study the properties of observed returns, 1-step unsmoothed returns, and 3-step unsmoothed returns. Columns with "+Dimson" also apply the Dimson (1979) method (with three lags), and columns with "+CDimson" also apply the constrained Dimson method (see Footnote 20). We report the average results obtained from 1,000 simulations of this panel of funds. See Section 1 for unsmoothing

Figure 4: Table 5B: MA(3) simulation with eight-factor model



(a) Figure 2: liquidity groups

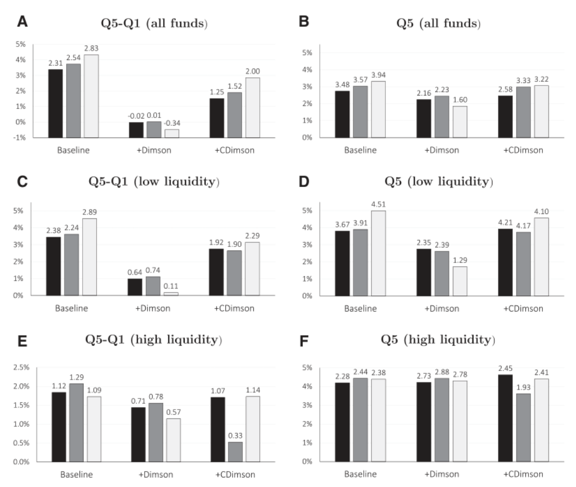


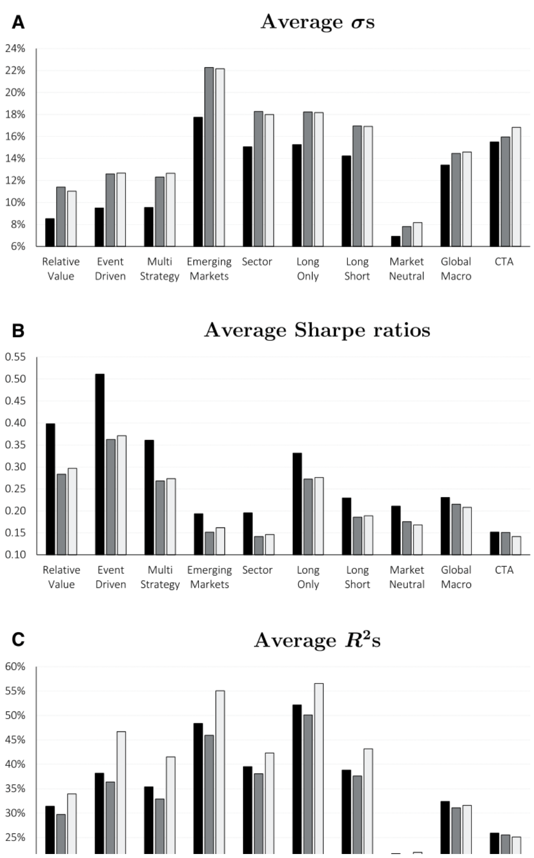
Figure 3
αs of hedge fund quintile portfolios formed based on the t-stat of past αs
The figure plots αs (with their t_{stat} on the top of each bar) of quintile portfolios formed by sorting hedge fund
based on the t-stat of their past αs (on a 24-month rolling window) measured using observed returns, 1-ste

(b) Figure 3: alpha portfolios

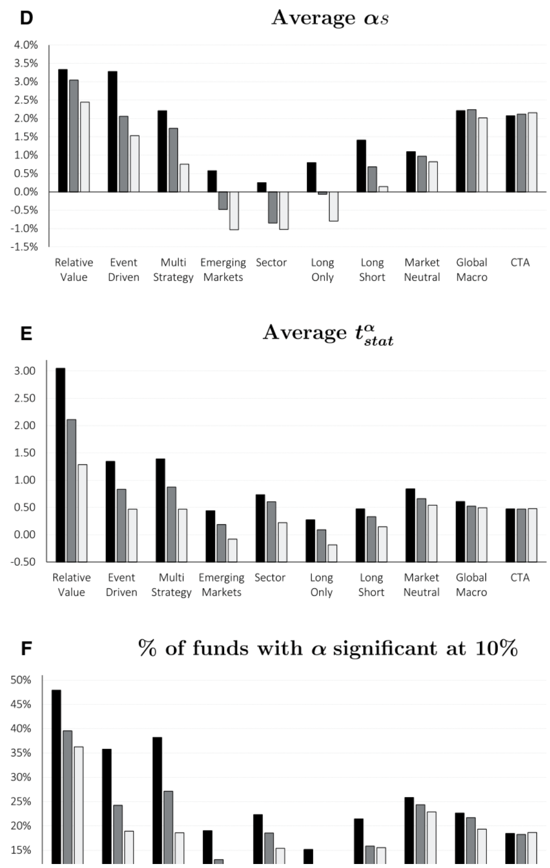
many strategies.

- Figure 5 reports the size of changes from observed to 1-step and from 1-step to 3-step returns.
- Figure 6 decomposes R^2 by Fung–Hsieh factors.

Economic message. The higher R^2 after 3-step unsmoothing reflects stronger measured exposure to identifiable systematic risk factors.



(a) Figure 4A–C



(b) Figure 4D–F

5.6 Tables 6 and 7: private CRE funds

Read:

- Table 6 shows fund-level and aggregate CRE autocorrelations. 3-step AR unsmoothing reduces aggregate autocorrelation substantially.
- Table 7 shows CRE volatility rises after unsmoothing, while alpha falls from about 4.0% to 0.8% in the two-factor model and R^2 rises from 4.6% to 16.0%.

Economic message. Private CRE funds look safer and more alpha-rich when systematic risk is smoothed.

6 Short summary

- Illiquid funds report smoothed returns because prices and valuations update with lags.

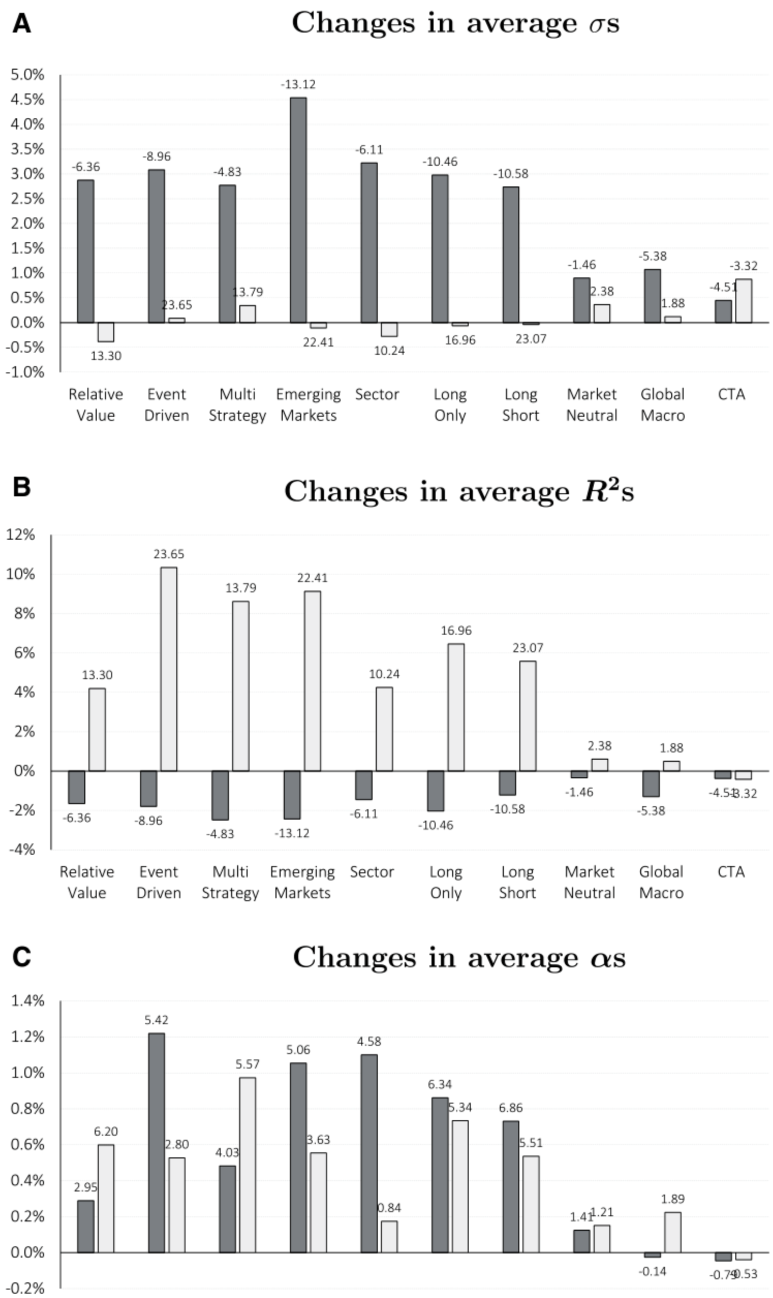
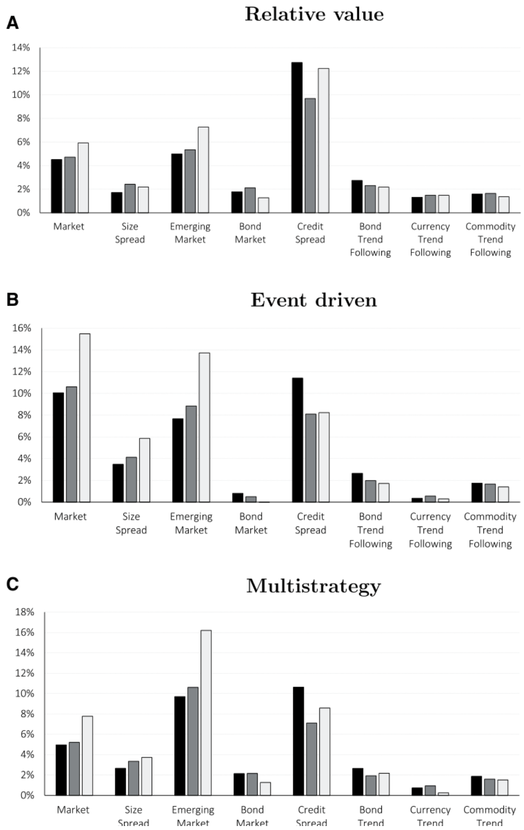
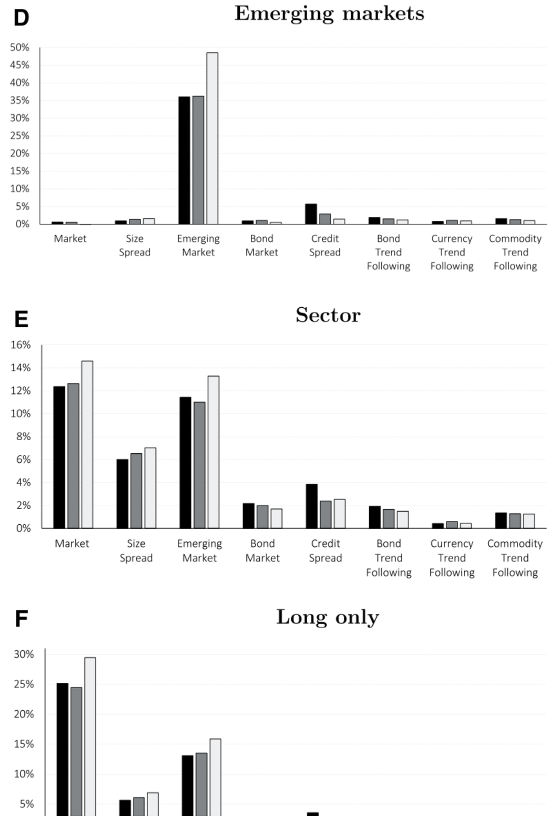


Figure 7: Figure 5: changes in risk and performance

- Smoothed returns generate autocorrelation and understate volatility, betas, and systematic risk.
- Traditional 1-step unsmoothing removes fund-level autocorrelation but can leave aggregate systematic autocorrelation.
- The paper proposes a 3-step method that separately unsmooths aggregate and relative components.
- 3-step unsmoothing raises factor-model R^2 and lowers alphas for illiquid hedge fund strategies.
- For private CRE funds, 3-step AR unsmoothing increases measured risk exposures and reduces average alpha toward zero.



(a) Figure 6A–C



(b) Figure 6D–F

7 Conclusion

7.1 Core conclusion

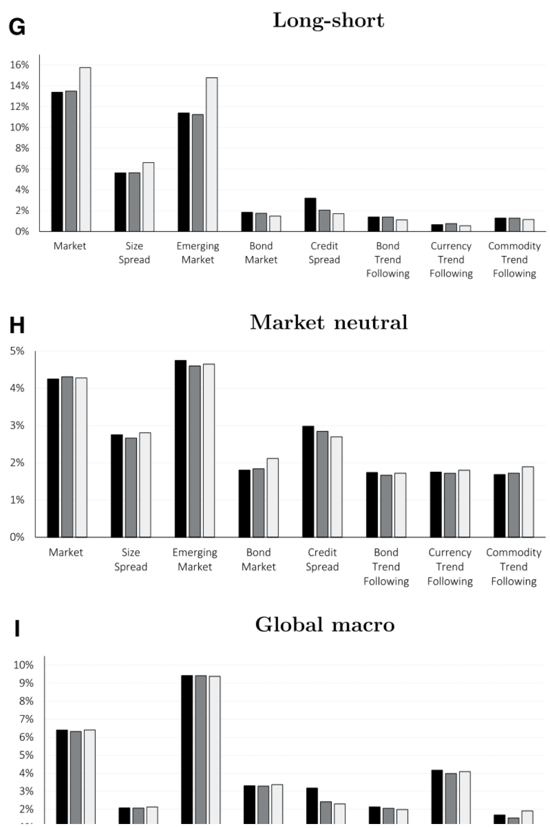
Couts, Gonçalves, and Rossi show that standard unsmoothing methods can leave systematic risk smoothed. Their 3-step procedure recovers returns that are less autocorrelated both at the fund level and when funds are aggregated into economically similar groups.

7.2 Economic intuition

Illiquid-fund managers report returns that reflect stale valuations. If all funds in a strategy share a stale common risk exposure, unsmoothing each fund in isolation can remove idiosyncratic autocorrelation while missing the common component. Aggregating funds reveals the leftover autocorrelation.

7.3 Takeaway

Illiquid fund performance is often lower than it looks. Better unsmoothing raises measured risk, increases R^2, and reduces alpha, especially for the most illiquid strategies and private real estate funds.



(a) Figure 6G–I

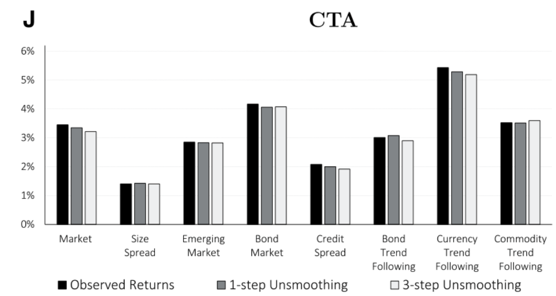


Figure 6

(b) Figure 6J

Table 6
Autocorrelations of private CRE fund returns

	Returns	Autocorrelations Cor_1	Cor_2	Cor_3	Cor_4	Partial autocorrelations Cor_1	Cor_2	Cor_3	Cor_4
Fund level	**Observed**	0.45 (63.6%)	0.39 (74.2%)	0.23 (51.5%)	0.21 (40.9%)	0.45 (59.1%)	0.10 (21.2%)	0.01 (9.1%)	0.01 (12.1%)
	1-step	−0.03 (0.0%)	0.02 (12.1%)	0.04 (7.6%)	0.10 (21.2%)	−0.02 (0.0%)	0.03 (12.1%)	0.05 (7.6%)	0.08 (21.2%)
	3-step	−0.04 (0.0%)	0.13 (27.3%)	−0.05 (0.0%)	0.17 (31.8%)	−0.02 (0.0%)	0.11 (12.1%)	−0.04 (0.0%)	0.14 (21.2%)
Aggregate level	**Observed**	0.75 (0.0%)	0.67 (0.0%)	0.42 (0.0%)	0.31 (0.3%)	0.65 (0.0%)	0.42 (0.1%)	−0.33 (0.9%)	0.00 (99.7%)
	1-step	0.46 (0.0%)	0.31 (0.3%)	0.12 (26.6%)	0.07 (48.3%)	0.41 (0.0%)	0.16 (18.3%)	−0.09 (45.0%)	0.02 (86.8%)
	3-step	−0.02 (86.7%)	0.24 (2.3%)	−0.09 (37.7%)	0.19 (6.9%)	0.02 (84.2%)	0.20 (6.0%)	−0.09 (38.5%)	0.14 (18.5%)

The table reports (average fund-level and aggregate) autocorrelations (from 1 to 4 quarters) for U.S. private commercial real estate (CRE) funds. Autocorrelations are based on observed returns, 1-step unsmoothed returns (as in Geltner 1991, 1993), and 3-step unsmoothed returns. In the upper panel, the numbers in parentheses reflect the fraction of funds with the respective autocorrelation being significant at 10% level. In the lower panel, the numbers in parentheses reflect the p-value for the test of whether the respective autocorrelation differs from zero. Partial autocorrelations refer to coefficients from a multivariate regression that includes lagged returns from up to 4 quarters. The sample goes from Q1 1994 through Q4 2017 and is restricted to private CRE funds that report return data to NCREIF and have at least 36 quarterly observations. See Subsection 3.1 for the AR unsmoothing methods used and Subsection 3.2 for further empirical details.

aggregate level, with the aggregate private CRE returns displaying a 1-quarter

(a) Table 6: CRE autocorrelations

Table 7
Risk and performance of private CRE funds

Statistics are related to	Raw performance $\mathbb{E}[r]$	σ	$\mathbb{E}[r]/\sigma$	1-factor model α	β_{re}	R^2	2-factor model α	β_{re}	β_e	R^2
Observed, R_o	5.0%	13.1%	0.38	4.3% (53.0%)	0.07 (27.3%)	3.2%	4.0% (51.5%)	0.02 (9.1%)	0.10 (7.6%)	4.6%
1-step, R_{1s}	5.0%	25.3%	0.20	2.7% (13.6%)	0.22 (47.0%)	7.9%	2.4% (13.6%)	0.15 (30.3%)	0.15 (10.6%)	9.9%
3-step, R_{3s}	5.0%	24.1%	0.21	1.6% (9.1%)	0.34 (89.4%)	13.7%	0.8% (9.1%)	0.21 (30.3%)	0.26 (13.6%)	16.0%
From R_o to R_{1s}	0.0%	12.1%	−0.18	−1.6% [−8.71]	0.16 [8.40]	4.8%	−1.7% [−7.36]	0.13 [6.44]	0.05 [1.62]	5.3%
From R_{1s} to R_{3s}	0.0%	−1.1%	0.01	−1.2% [−3.93]	0.12 [3.70]	5.7%	−1.6% [−4.42]	0.06 [1.95]	0.12 [4.03]	6.0%
From R_o to R_{3s}	0.0%	11.0%	−0.17	−2.7% [−11.73]	0.27 [11.01]	10.5%	−3.2% [−11.34]	0.19 [9.62]	0.17 [8.38]	11.3%

The table reports (average fund-level) statistics related to the risk and performance of U.S. private commercia real estate (CRE) funds. All statistics are based on observed returns, 1-step unsmoothed returns (as in Geltne 1991, 1993), and 3-step unsmoothed returns. The upper panel reports the values of the statistics (with the % o funds with significant values at 10% in parentheses) and the lower panel reports changes in these statistics (wit the t_{stat} for a test of whether the mean change differs from zero in brackets). The sample goes from Q1 199 through Q4 2017 and is restricted to private CRE funds that report return data to NCREIF and have at least 3 quarterly observations. See Subsection 3.1 for the AR unsmoothing methods used and Subsection 3.2 for furthe empirical details.

(b) Table 7: CRE risk and performance